

Grey-Box Modeling for Predictive Irrigation Control in Crop Systems

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Abstract—This paper presents the design and implementation of a Model Predictive Controller (MPC) for regulating soil moisture in the root zone of a crop. A simplified dynamic model based on the soil water balance is employed, incorporating crop evapotranspiration and precipitation as external disturbances, as well as a moisture-dependent deep drainage term. The controller is formulated as a non-linear optimization problem with physical and operational constraints. Additionally, a preliminary grey-box approach based on a simple neural approximation is explored, aimed at evaluating its potential future integration as a complement to the physical model. Simulation results demonstrate that the MPC is capable of maintaining soil moisture close to a reference value, anticipating climatic disturbances and avoiding soil over-saturation.

Index Terms—automatic irrigation, CasADI, deep drainage, evapotranspiration, predictive control, water balance.

I. INTRODUCTION

Efficient water management in agriculture has become a critical challenge under increasing water scarcity, climate variability, and growing food demand. Irrigated agriculture accounts for approximately 70% of global freshwater withdrawals and contributes around 40% of total food production while occupying less than 20% of cultivated land [1], [2]. Projections indicate that global food demand may increase by more than 60% by 2050, intensifying pressure on limited water resources [3], [4]. Consequently, improving irrigation efficiency is essential to ensure food security and environmental sustainability.

Soil moisture in the root zone directly determines crop growth, yield formation, and physiological stability. Water deficit conditions induce plant stress, reduce photosynthetic activity, and decrease productivity, while excessive moisture promotes leaching, nutrient losses, and waterlogging [5], [6]. Maintaining soil moisture within agronomically optimal bounds is therefore fundamental for maximizing water use efficiency and minimizing environmental impacts [4], [7]. These considerations motivate the development of advanced irrigation management strategies capable of dynamically regulating the soil water balance.

Traditional irrigation scheduling is typically based on fixed calendars, empirical rules, or simplified evapotranspiration estimates without real-time soil feedback [8], [9]. Although widely adopted, such open-loop strategies do not explicitly account for the nonlinear dynamics of the soil–plant–atmosphere

system nor for stochastic climatic disturbances [10]. As a consequence, these strategies frequently result in either over-irrigation or deficit irrigation, decreasing water productivity and potentially contributing to soil degradation processes [4]. These limitations highlight the need for closed-loop control approaches grounded in dynamic system modeling.

The FAO-56 guideline provides a standardized and globally accepted methodology for estimating crop evapotranspiration using the Penman–Monteith equation [11]. Despite its robustness for estimating crop water requirements, the conventional application of FAO-56 in irrigation scheduling is typically implemented in an open-loop manner, without dynamic feedback from real-time soil moisture measurements. As a result, it does not inherently compensate for model uncertainties, parameter variability, or unexpected climatic disturbances. Embedding FAO-based evapotranspiration models within closed-loop control frameworks therefore represents a promising approach to enhance irrigation performance and system resilience [12].

While closed-loop strategies such as moisture threshold methods have been traditionally used, more advanced decision support systems (DSS) have been proposed to manage the operational complexity of irrigation management [13]. Although these approaches improve responsiveness compared to open-loop schemes, they present limitations in handling multivariable constraints, time delays, and predictive disturbance rejection [14]. Moreover, soil moisture dynamics are inherently nonlinear and subject to uncertain external disturbances such as precipitation and evapotranspiration variability [15]. Therefore, more advanced control methodologies are required to effectively address these challenges.

Model Predictive Control (MPC) has emerged as a powerful framework for agricultural water management due to its ability to incorporate dynamic models, constraints, and future disturbance predictions into an optimization-based decision process [16]–[18]. Recent studies demonstrate that MPC-based irrigation strategies can significantly reduce water consumption while maintaining or improving crop yield compared to conventional methods [14], [15]. By solving a receding-horizon optimization problem, MPC anticipates climatic disturbances and enforces physical constraints such as soil saturation limits and actuator bounds. These features make MPC particularly suitable for regulating soil moisture within agronomic and hydrological constraints.

In addition to purely physics-based modeling, hybrid Grey-Box approaches that combine mechanistic models with data-driven techniques have gained increasing attention in agricul-

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tural water management [7], [19]. Machine learning models, particularly neural networks, can be employed to capture residual nonlinear dynamics not explicitly represented in simplified soil water balance formulations. By complementing physical knowledge with data-driven corrections, these hybrid strategies aim to improve predictive performance while maintaining a degree of physical interpretability. Such integration is consistent with emerging trends toward intelligent and explainable agricultural control systems.

This work proposes a predictive irrigation control scheme based on a dynamic soil water balance model that explicitly incorporates crop evapotranspiration, precipitation disturbances, and moisture-dependent deep drainage. Furthermore, a preliminary Grey-Box extension based on a residual neural network is explored to compensate for unmodeled dynamics while preserving physical interpretability. The remainder of this article is organized as follows: Section II presents the system modeling framework; Section III details the predictive control formulation and implementation; Section IV discusses simulation results under different operational scenarios; and Section V provides a comprehensive validation of the Grey-Box model along with concluding remarks.

II. DYNAMIC SYSTEM MODEL

This section describes the soil–plant–atmosphere dynamics based on the water flows of irrigation, precipitation, evapotranspiration, and drainage. The root zone is modeled through a water balance equation where volumetric soil moisture θ serves as the state variable. Following the classical approach in [11], the continuous dynamics are expressed as:

$$\frac{d\theta}{dt} = \frac{1}{Z_r} (I(t) + P(t) - ET_c(t, \theta) - D(t, \theta)), \quad (1)$$

where Z_r is the effective root depth (mm), $I(t)$ and $P(t)$ represent irrigation and precipitation, $ET_c(t, \theta)$ is crop evapotranspiration, and $D(\theta)$ is deep drainage. This formulation provides the mathematical foundation for the predictive controller developed in Section III.

A. Precipitation modeling

Precipitation events are modeled using a continuous double sigmoid formulation to realistically represent rainfall growth and decay. This approach avoids the temporal discontinuities of traditional step functions, facilitating the numerical stability of the optimization solver. The rain intensity $P(t)$ is defined as

$$P(t) = P_{\max} \left[\frac{1}{1 + e^{-a(t-t_1)}} - \frac{1}{1 + e^{-a(t-t_2)}} \right],$$

where $P_{\max}$ is the maximum intensity, a is the smoothing parameter, and t_1, t_2 delimit the event duration. Such a smooth profile accurately reproduces natural precipitation dynamics while maintaining the differentiability required for the nonlinear MPC.

B. Evapotranspiration and Drainage Dynamics

Actual crop evapotranspiration $ET_c(t, \theta)$ is modeled by scaling the potential atmospheric demand with soil water availability. It is defined as $ET_c(t, \theta) = ET_{c,\text{pot}}(t)K_s(\theta)$, where $ET_{c,\text{pot}}(t)$ represents the potential evapotranspiration derived from climatic variables and the Penman-Monteith equation. The water stress coefficient $K_s(\theta)$ varies linearly between the wilting point θ_{wp} and field capacity θ_{fc} to represent the physiological reduction of transpiration. This continuous formulation ensures that the evapotranspiration feedback remains differentiable, avoiding the numerical instabilities associated with discrete threshold models.

Deep drainage $D(\theta)$ represents water loss below the root zone when moisture exceeds field capacity. To maintain numerical stability in the MPC solver, this process is modeled using a smooth *softplus* function: $D(\theta) = K_d \frac{1}{\alpha} \ln(1 + e^{\alpha(\theta - \theta_{fc})})$. The smoothing parameter α dictates the transition sharpness as moisture exceeds field capacity, ensuring C^2 continuity for the optimization algorithm. This continuous approximation effectively mimics gravitational percolation while providing C^2 continuity for the optimization algorithm. The resulting mathematical model establishes a robust foundation for the predictive control strategy and the optimization problem formulated in Section III.

III. CONTROL METHODOLOGY

This section describes the predictive controller designed to regulate soil moisture θ while optimizing water resources. As illustrated in Fig. 1, the MPC utilizes the dynamic water balance model to predict the system's evolution and calculate the optimal irrigation $I(t)$ under climatic disturbances. This optimization-based approach directly incorporates the differentiable formulations of evapotranspiration and drainage defined in Section II. Consequently, the controller ensures operational stability by anticipating future state trajectories within a constrained environment.

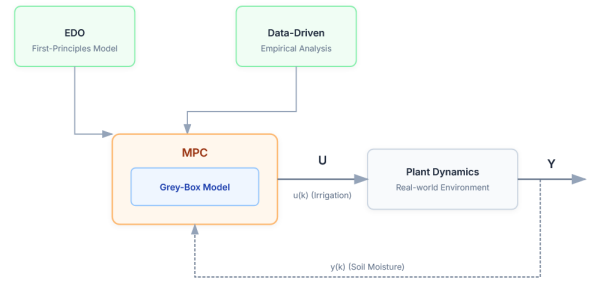


Figure 1. Block diagram of the irrigation control system.

A. Model Predictive Control Formulation

The water balance (1) is discretized using the forward Euler method with a sampling step dt to enable recursive modeling. This discrete-time formulation expresses the soil moisture $\theta(k)$ as a function of irrigation $I(k)$ and disturbances, facilitating efficient numerical optimization. The controller's performance

is governed by an objective function that penalizes both the moisture deviation from a reference θ_{ref} and excessive water use. Following standard MPC formulations [16], the cost function J is defined as

$$J = \sum_{k=1}^N (\theta(k) - \theta_{\text{ref}})^2 + \lambda I(k)^2, \quad (2)$$

where N is the prediction horizon and λ balances regulation precision against resource conservation.

To ensure physically viable trajectories, the MPC incorporates constraints on state and control variables throughout the horizon. Soil moisture is bounded by $\theta_{\min} \leq \theta(k) \leq \theta_{\max}$, with limits set at 0.1 and 0.4, respectively, while irrigation is restricted to $0 \leq I(k) \leq I_{\max}$. The inclusion of integral action within this constrained framework compensates for persistent modeling errors without leading to signal saturation. This structure ensures robust steady-state performance and prevents soil over-saturation during intense precipitation events.

B. Grey-Box Modeling Approach

The Grey-box framework integrates deterministic physical knowledge with the adaptive flexibility of data-driven methods. In this work, the system dynamics are decomposed into a nominal physical component and a learned residual term, expressed as

$$\theta_{GB}(k+1) = f_{\text{phys}}(\theta(k), u(k), d(k)) + \text{ANN}(x_{\text{in}}(k)),$$

where f_{phys} represents the discretized water balance and ANN denotes the residual $r(k)$. The neural network utilizes a Multi-layer Perceptron architecture with two hidden layers and $\tanh$ activation functions to ensure a smooth, differentiable mapping. This input vector $x_{\text{in}}(k) = [\theta(k), u(k), P(k), ET_c(k)]^T$ incorporates the state, control action, and disturbances for comprehensive residual identification.

The learning objective focuses on the discrepancy $\Delta\theta = \theta_{\text{real}} - \theta_{\text{model}}$ to compensate for unmodeled phenomena such as non-linear drainage or varying crop coefficients. By targeting this residual, the network enhances predictive accuracy without losing the physical interpretability of the irrigation input. This structure provides a high-fidelity prediction horizon that combines conservation laws with adaptive machine learning corrections. The performance of this hybrid control architecture is evaluated through the simulation scenarios and the grey-box identification tests presented in Section IV.

IV. SIMULATION RESULTS

This section evaluates the Model Predictive Controller (MPC) performance across various operating scenarios using a daily time step ($dt = 1$). All numerical simulations were implemented in MATLAB R2025 and the CasADi framework on an AMD Ryzen 5 7520U system with 16 GB of RAM. This setup ensured efficient execution of the NLP solver (IPOPT), with average computation times per iteration suitable for real-time irrigation scheduling. The complete implementation and scripts are available in a public repository¹.

¹<https://github.com/MiguelPanqueva2023/soil-moisture-mpc-greybox>

A. Incremental Model Evolution and Controller Response

The validation followed an incremental refinement process to enhance the physical fidelity of the soil-plant system representation. Initial tests utilized idealized storage dynamics and rectangular pulse disturbances, which were later replaced by smooth double sigmoid functions to represent realistic precipitation events. As shown in Figs. 2, 3, and 4, the MPC demonstrates high anticipatory capacity by adjusting irrigation before water inputs and compensating for time-varying evapotranspiration. This evolution from discrete to continuous modeling was essential to ensure the differentiability required by the nonlinear optimizer.

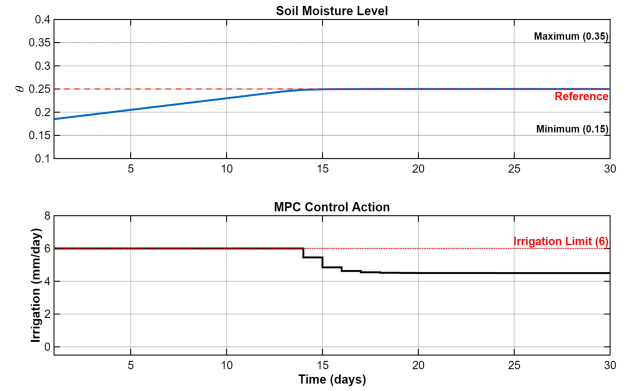


Figure 2. Evolution of soil moisture and irrigation signal for the base model.

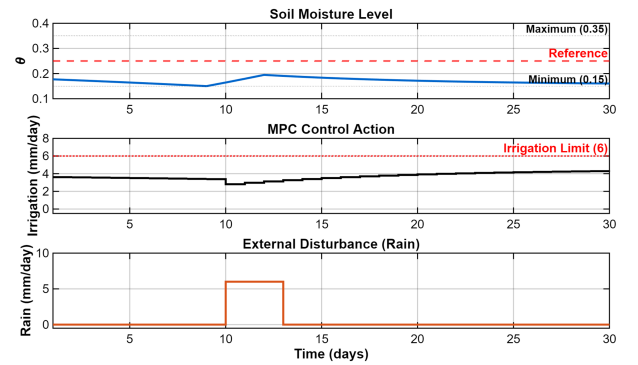


Figure 3. Comparison of the irrigation signal in the presence and absence of precipitation.

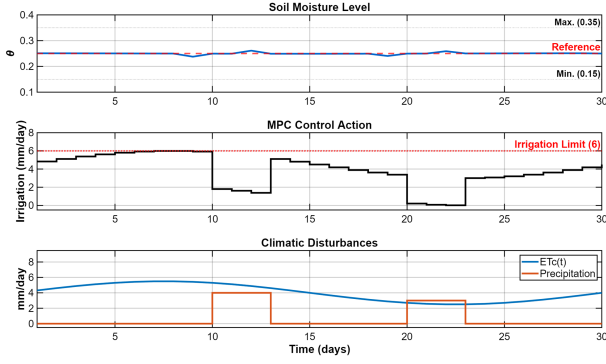


Figure 4. Time-varying crop evapotranspiration and irrigation response obtained via MPC.

Biophysical complexity was further increased by incorporating moisture-dependent drainage and the water stress coefficient (K_s). The implementation of *sofplus* smoothing and integral action effectively eliminated steady-state errors while preventing numerical chattering in the derivatives. As illustrated in Figs. 5 and 6, these continuous formulations allow the controller to maintain stability across the entire moisture range. These results evidence a rich balance dynamic where losses adjust to the actual water availability in the soil profile.

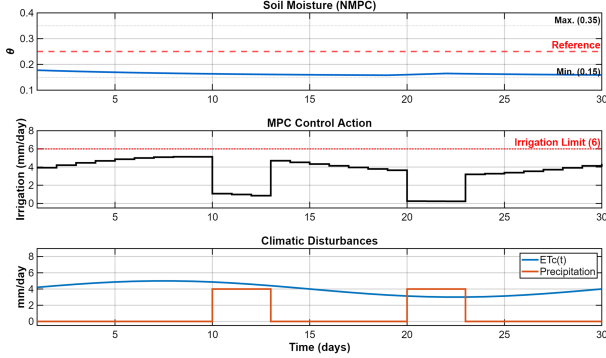


Figure 5. Water stress coefficient, moisture-dependent drainage, and soil moisture evolution.

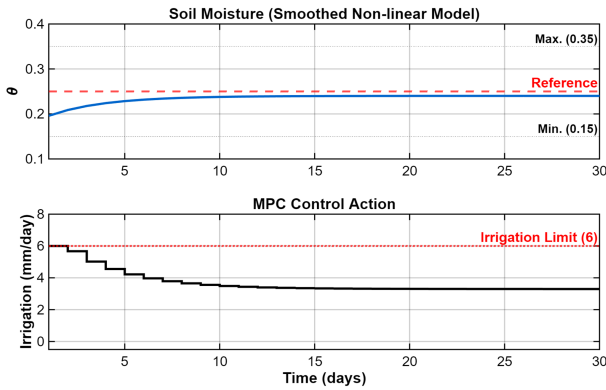


Figure 6. MPC response under smooth formulations for drainage and water stress (K_s).

The final approach included "smooth physical saturations" to handle moisture and control limits without introducing discontinuities. As shown in Fig. 7, this stage was crucial for softening the controller's response to the system's physical constraints. In the comprehensive 60-day climatic scenario (Fig. 8), the system responds robustly to variable moisture references and realistic sigmoidal disturbances. This performance validates a control scheme that successfully balances agronomic precision with numerical stability.

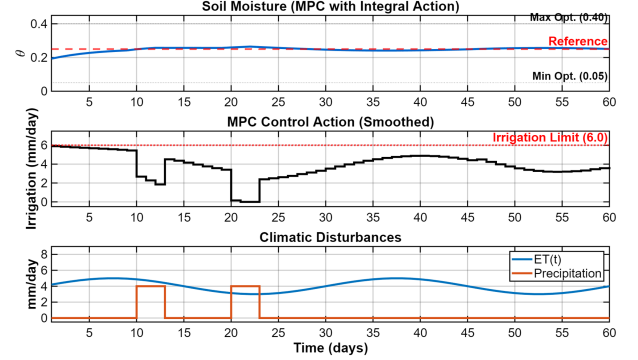


Figure 7. Effect of smooth physical saturations and integral action on soil moisture stability.

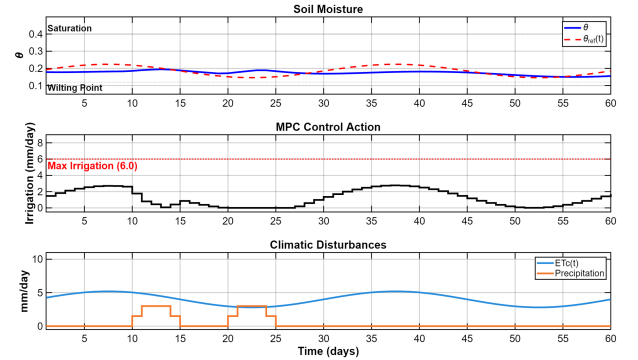


Figure 8. MPC performance under variable moisture references and contrasting climatic scenarios.

B. Grey-Box Identification: Data Generation and Training

A Grey-Box identification process was implemented to bridge the gap between theoretical laws and unmodeled biophysical dynamics. The residual $\Delta\theta = \theta_{real} - \theta_{model}$ was targeted using synthetic data to capture systematic biases like non-linear percolation and varying crop coefficients. As shown in Fig. 9, these residuals exhibit learnable temporal structures rather than purely stochastic noise. This predictability confirms that data-driven corrections can effectively complement the nominal physical model to enhance accuracy.

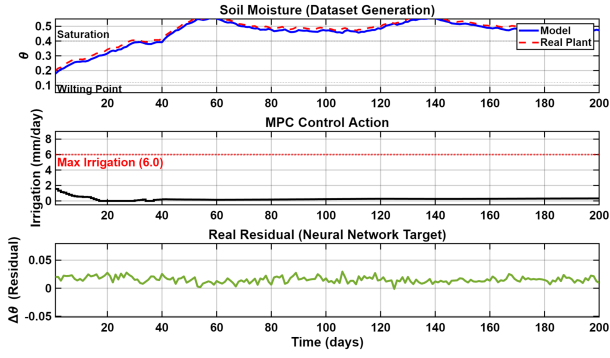


Figure 9. Synthetic data generation: moisture evolution, irrigation, and residual term target.

The residual identification utilized the MLP architecture described in Section III, trained over 3,000 epochs with a learning rate of 0.01. Input normalization and *tanh* activation functions ensured a smooth mapping that accurately tracks the residual trend without introducing instabilities. As illustrated in Fig. 10, the trained model estimates the full numerical range of discrepancies with high fidelity. This configuration prepares the system for real-time closed-loop operation while preserving the physical interpretability of the control action.

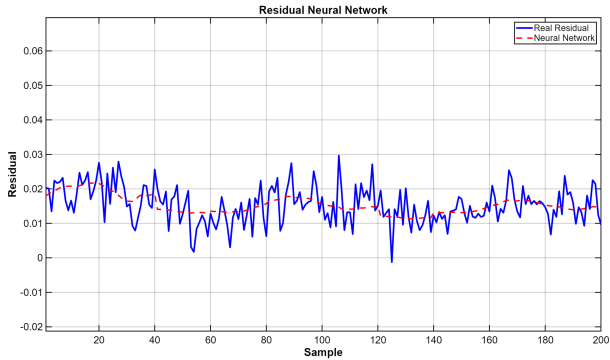


Figure 10. Comparison between the system's actual residual and the residual estimated by the ANN.

C. Integration of the Grey-Box model into the MPC

The residual neural network provides a corrective term $r(k)$ that scales the nominal physical prediction in the form $x_{k+1} = \theta_{phys} + r$. This hybrid approach allows the controller to compensate for systematic modeling gaps and unmodeled drainage effects in real-time. As demonstrated in Fig. 11, the Grey-Box scheme significantly improves reference tracking compared to the purely physical model. This integration enhances system resilience without increasing control effort, fulfilling the objective of high-fidelity agricultural regulation.

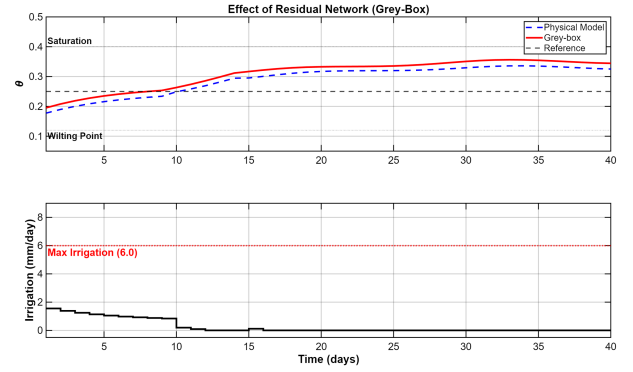


Figure 11. Comparison between the physical model and the Grey-Box scheme within the MPC.

V. CONCLUSIONS AND FINAL REMARKS

This study demonstrates that incorporating physical complexity, such as moisture-dependent drainage and water stress, significantly enhances soil dynamics representation within a predictive framework. While initial models faced numerical challenges, the implementation of *softplus* smoothing functions ensured the convergence of the nonlinear optimizer without compromising physical consistency. These results confirm that a refined water balance model provides a robust foundation for stable irrigation regulation across diverse climatic scenarios. Consequently, the proposed architecture effectively balances agronomic precision with the computational requirements of real-time control systems.

The transition to a hybrid Grey-Box approach successfully bridged the gap between simplified physical laws and complex unmodeled dynamics. By utilizing a residual neural network to compensate for systematic biases, the controller achieved superior reference tracking and enhanced predictive accuracy. This dual methodology proves that combining mechanistic interpretability with machine learning flexibility offers a scalable solution for precision agriculture. Future work will focus on validating this scheme with experimental field data and assessing its performance under significant sensor noise and parametric uncertainty.

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